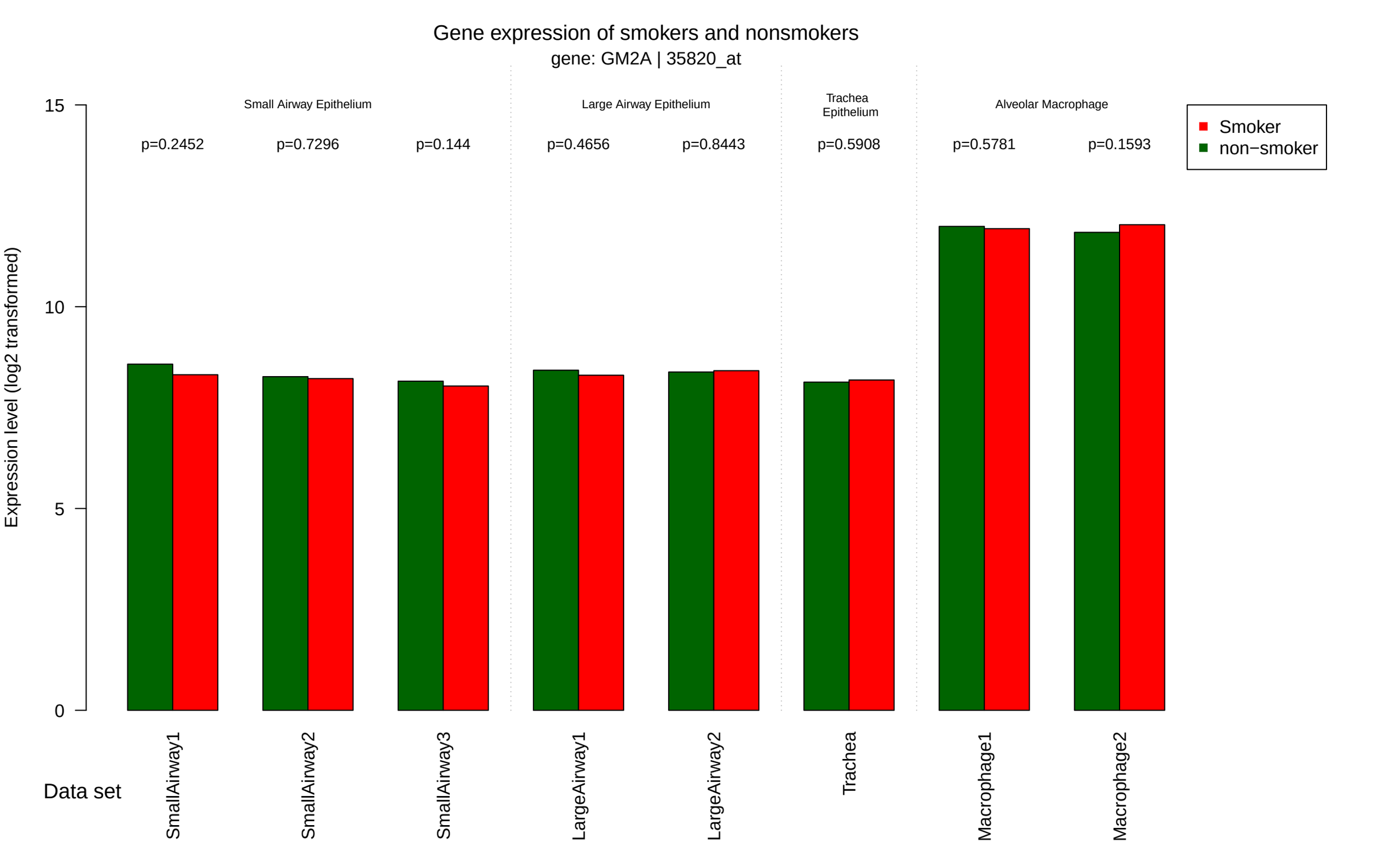
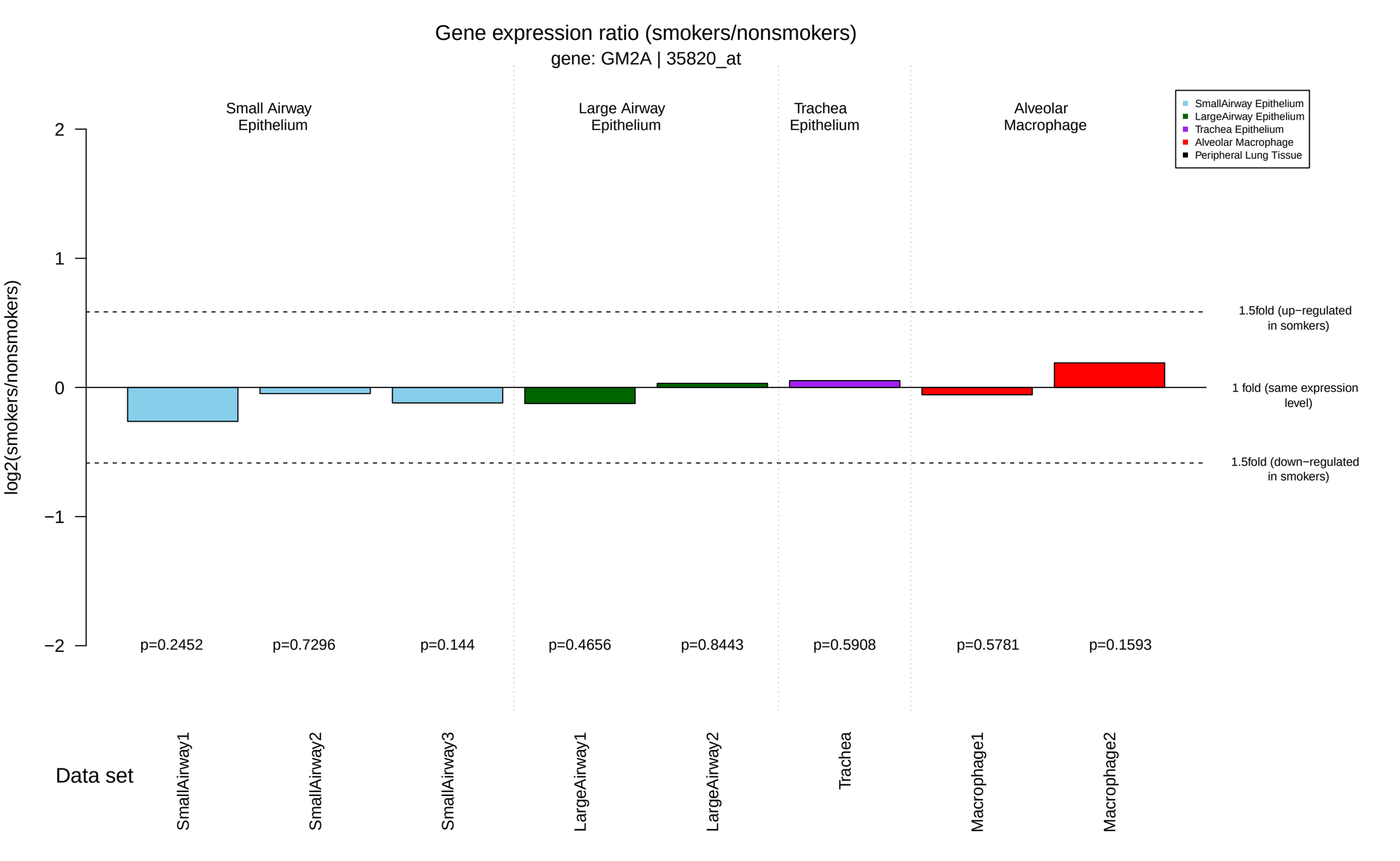
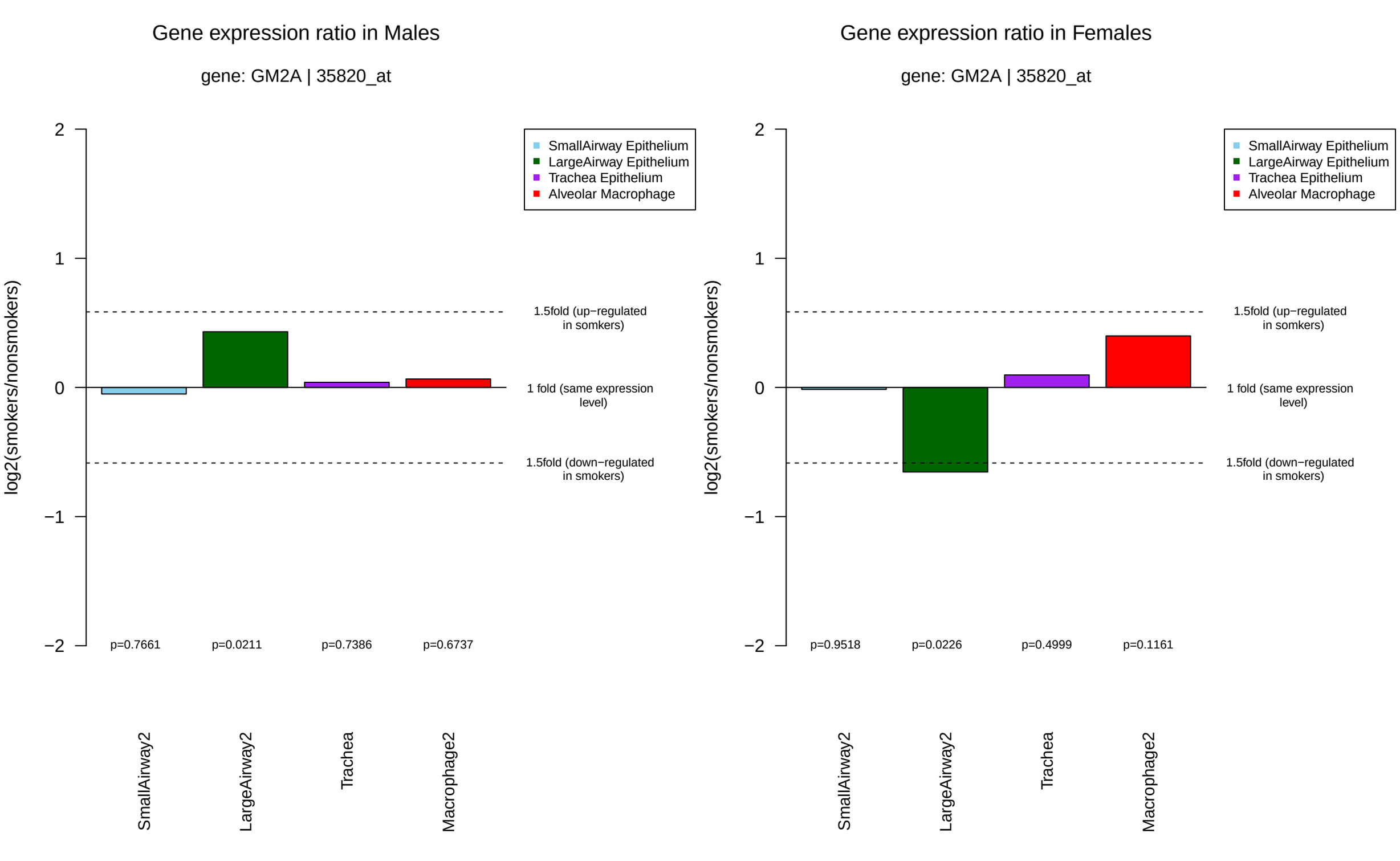
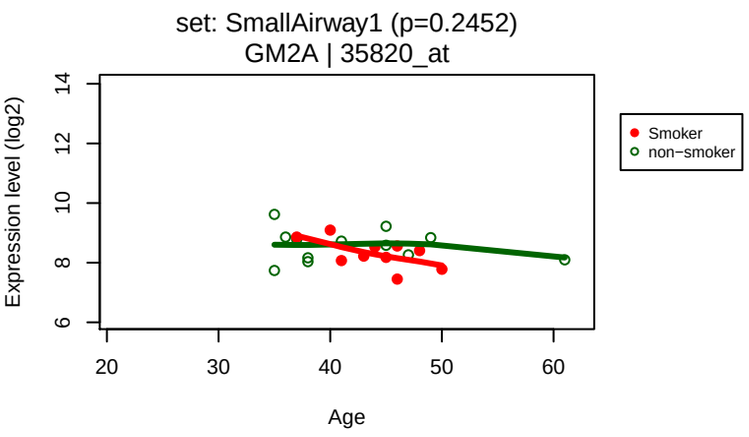
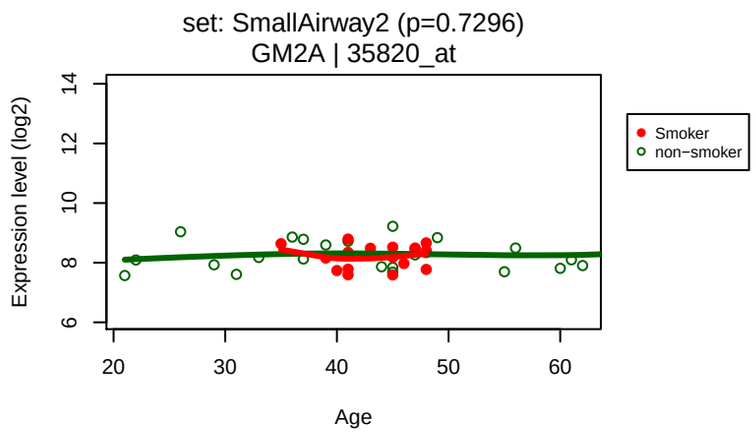
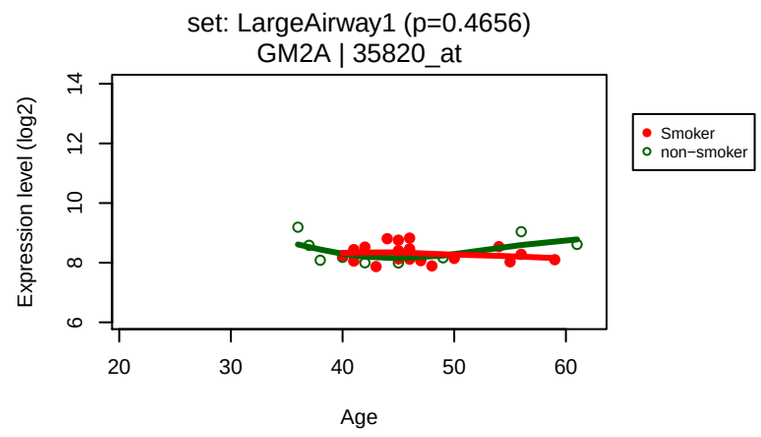

Gene expression of smokers and nonsmokers
gene: GM2A | 35820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2452
p=0.7296
p=0.144
p=0.4656
p=0.8443
p=0.5908
p=0.5781
p=0.1593
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

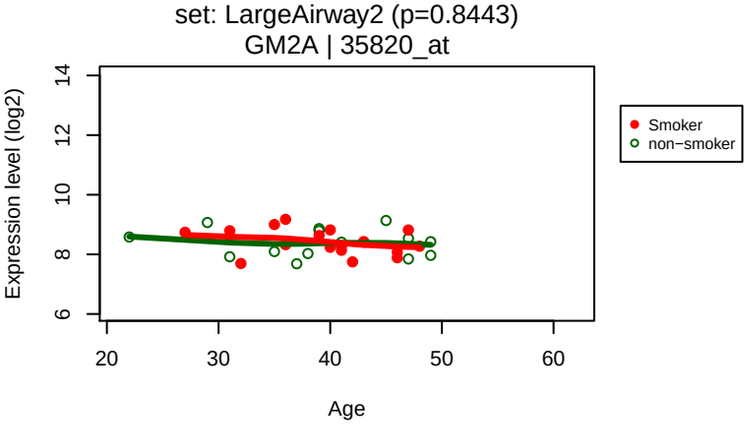

Gene expression ratio (smokers/nonsmokers)
gene: GM2A | 35820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2452
p=0.7296
p=0.144
p=0.4656
p=0.8443
p=0.5908
p=0.5781
p=0.1593
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

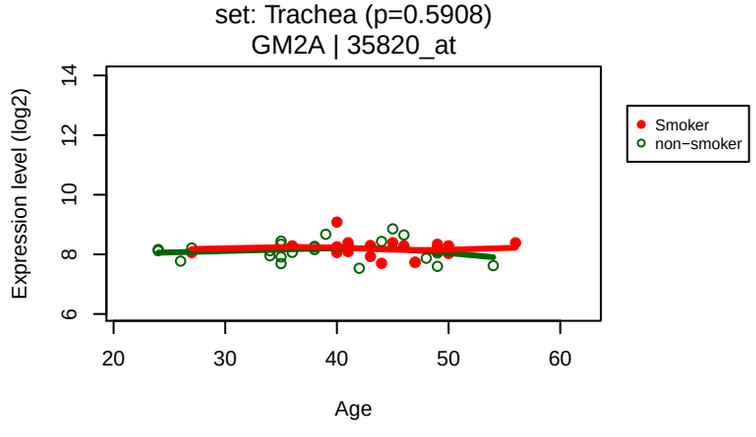

Gene expression ratio in Males
gene: GM2A | 35820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7661
p=0.0211
p=0.7386
p=0.6737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GM2A | 35820_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9518
p=0.0226
p=0.4999
p=0.1161
SmallAirway2
LargeAirway2
Trachea
Macrophage2

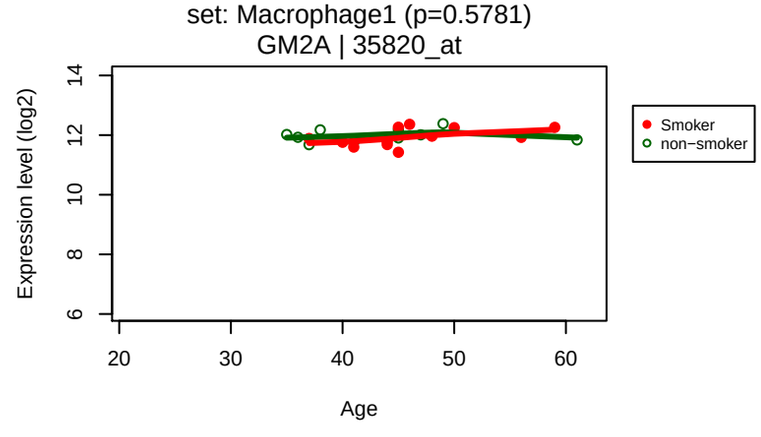

set: SmallAirway1 (p=0.2452)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker

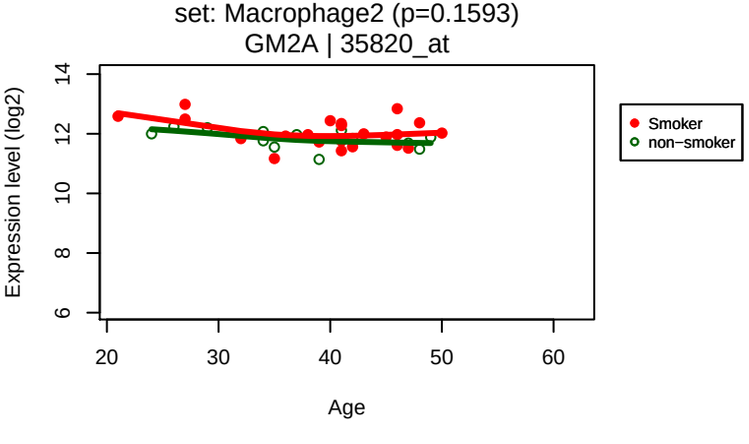

set: SmallAirway2 (p=0.7296)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker

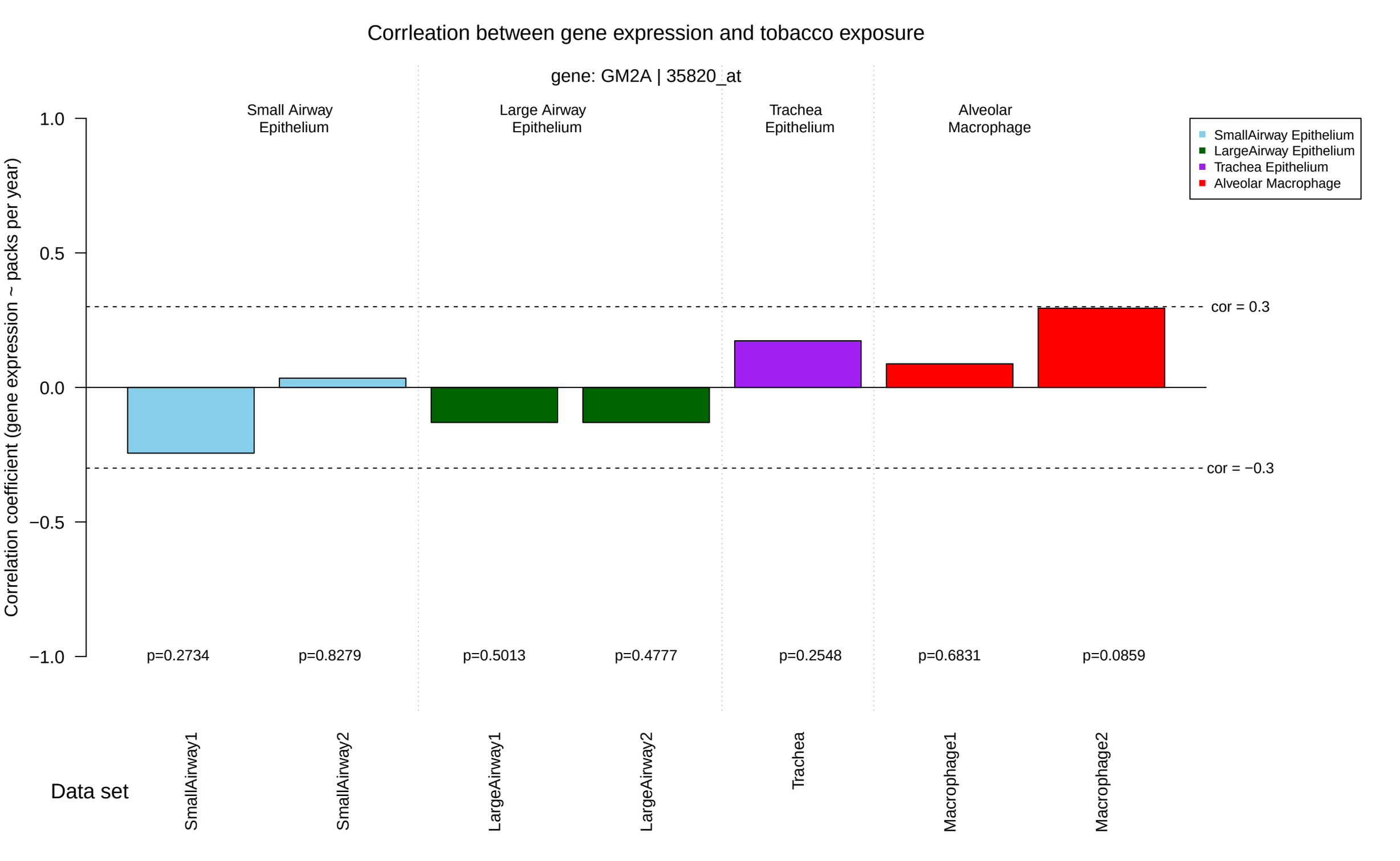

set: LargeAirway1 (p=0.4656)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8443)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5908)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5781)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1593)
GM2A | 35820_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GM2A | 35820_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.2734
p=0.8279
p=0.5013
p=0.4777
p=0.2548
p=0.6831
p=0.0859
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2